

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **PLC programming** and **SCADA system modifications** for production automation equipment, integrating **servo drives** and **vision systems** to enhance operational efficiency.
- Executed fault finding and troubleshooting on complex **automation equipment** within live production environments, diagnosing issues through interrogation of **PLC code** to ensure minimal disruption.
- Collaborated with cross-functional teams to develop and implement effective repairs and improvements for **robotics** and automated machinery, contributing to a **15% reduction** in unplanned downtime.
- Managed the technical design and selection of hardware and architecture for automation projects, ensuring adherence to **GAMP documentation** standards such as URS, FDS, FAT, and SAT.
- Analyzed opportunities for new or improved tools for process improvement activities, defining equipment and technologies based on expertise to drive a **20% increase** in machine productivity.
- Coached area technicians and junior control engineers on **automation troubleshooting** and problem-solving techniques, supporting continuous improvement efforts across the Biotools department.

SKILLS

Automation & Control – PLC, Robotics, Vision Systems, SCADA, Servo Drives, Motion Control, GAMP, URS, FDS, FAT, SAT, Machinery Directive, CE Compliance, EN60204-1, Electrical Control Panels

Programming & Scripting – PLC Programming (Siemens S7, Telemecanique, Rockwell), Python, Java, SQL

Networking & Protocols – EtherNet IP, DeviceNet, ControlNet, Modbus, Profibus

System Integration – MES, REST APIs, PostgreSQL, MySQL, Oracle

Project Management – Capital Projects, Budget Management, Requisitions, Vendor Contracts, Spares Management, Risk Assessment (RCAs), GMP, EHS, Plant Procedures

Tools & Technologies – Linux, Git, Jenkins, Jira, Confluence, Splunk, Dynatrace, AWS (EC2, S3, Lambda, CloudWatch), Docker

PROJECTS

Automated Production Line Optimization (HCL Technologies)

- Redesigned the control logic for a key production line using **Siemens S7 PLCs** and **EtherNet IP**, improving throughput by approximately **18%**.
- Developed diagnostic scripts in **Python** to analyze historical production data from **SCADA** logs, identifying recurring failure patterns in **servo motor** control loops.
- Implemented a new vision system inspection routine, reducing product defect rates by **30%** and improving quality control accuracy.

Manufacturing Execution System Integration (HCL Technologies)

- Integrated a legacy Java-based enterprise system with a new **Manufacturing Execution System (MES)** using **REST APIs** and direct **PostgreSQL** database connections.
- Automated data reconciliation workflows between the MES and the ERP system using **Python scripts**, reducing manual data entry errors by **~40%** per cycle.
- Developed standardized documentation and backup procedures for all control system configurations, ensuring compliance with internal audit requirements and facilitating quicker disaster recovery.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)